

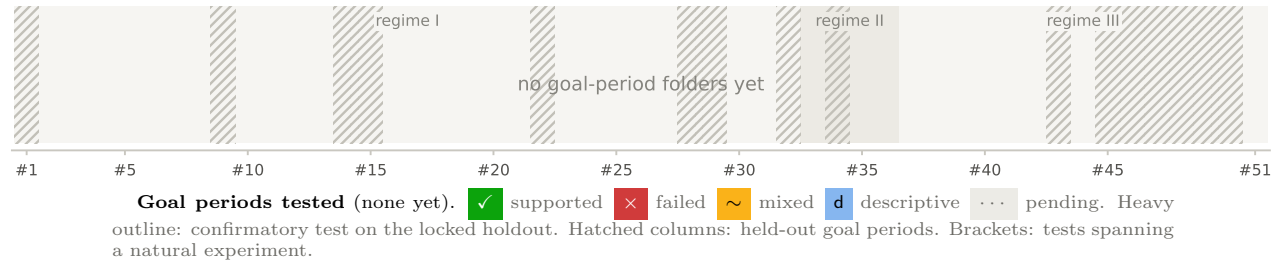
H06: Free weeks show neutral cooperative dynamics with a cooperator core

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

not yet rated

Bottom line. Pending.

level: hypothesis · pipeline checks 29% · updated —



WHAT HAPPENED

Not yet summarized. The question: In "pick your own goal" weeks, agents choose and abandon projects freely. Do project abundances follow *neutral cooperative dynamics* (Piñero-style: species, here projects, are equivalent, but joining is frequency-dependent and cooperative)? Signatures: bimodal abundances with a persistent cooperator core, Simpson concentration λ near the predicted $\lambda^*(\mu, N)$, and boundaries μ_B (bimodal) and μ_L (log-series) in the novelty rate μ . Or does plain Hubbell neutral drift (log-series, no cooperation) explain them? Practical payoff: whether a free swarm concentrates its effort on a few shared projects by a predictable law, and what sets how many projects survive.

PHYSICAL MODEL

From: physics-models/06-neutral-cooperative-dynamics (read its README for the rules, λ^* , μ_B , μ_L and the finite-N

simulation), with physics-models/10-potts as the categorical-state view. H11 found strong herding (ferromagnetic Potts coupling) on project labels in 11/14 periods, including free weeks #31 and #37. That is the cooperative ingredient this model formalizes.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.
Observables. Pending.